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Studierichting: Informatica
Oefeningenreeks 3C nr. 1

$$y^2 = 2x$$

$$x = \frac{y^2}{2}$$

$$d^2 = (x-1)^2 + (y-4)^2$$

$$d^2 = \left(\frac{y^2}{2} - 1\right)^2 + (y-4)^2$$

$$f(y) = \left(\frac{y^2}{2} - 1\right)^2 + (y-4)^2$$

$$f'(y) = 2\left(\frac{y^2}{2} - 1\right)\left(\frac{y^2}{2} - 1\right)' + 2(y-4)(y-4)'$$

$$\left(\frac{y^2}{2} - 1\right)' = y$$

$$(y-4)' = 1$$

$$f'(y) = 2\left(\frac{y^2}{2} - 1\right)y + 2(y-4)$$

$$f'(y) = y^3 - 2y + 2y - 8$$

$$f'(y) = y^3 - 8$$

$$f'(y) = 0$$

$$y^3 - 8 = 0$$

$$y = 2$$

$$x = \frac{y^2}{2}$$

$$x = \frac{4}{2}$$

$$x = 2$$

$$\boxed{\text{punt } (2,2)}$$